PRACTICAL SESSION · EVIDENCE-BASED PHARMACOLOGY

Searching the Literature for Practice Guidelines

Blood Module · Thrombotic Disease & Anemia

7 worked cases | PICO & PubMed | Guideline appraisal

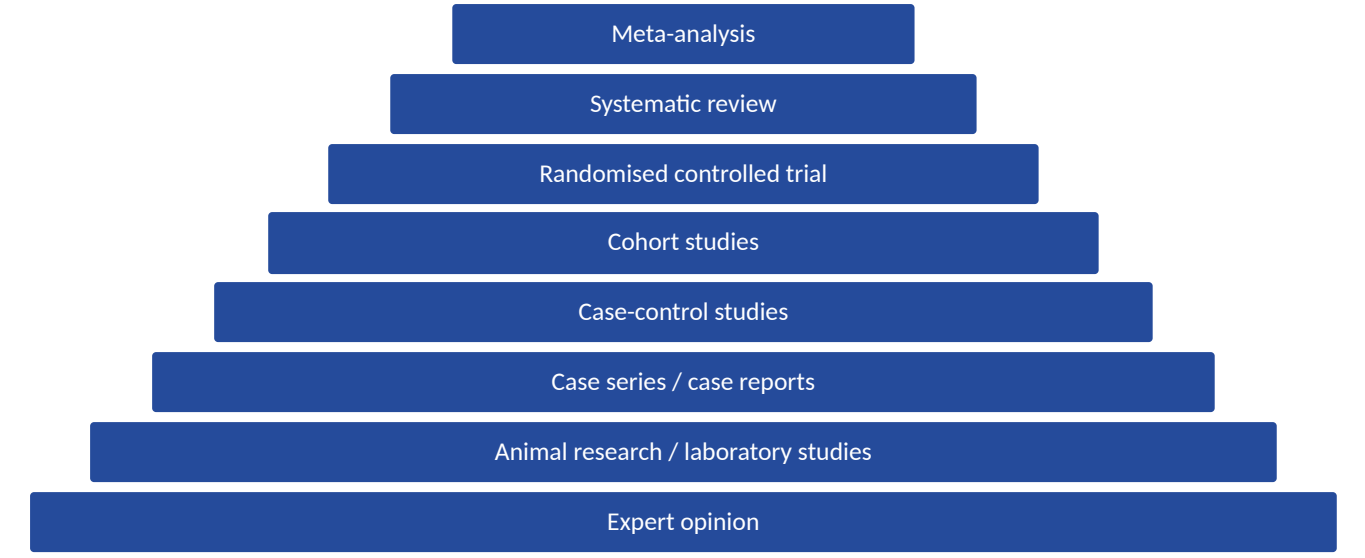
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Academic Year 2026 – 2027

WHERE RESEARCH FITS

Levels of evidence

Higher tiers pool and critically appraise many studies, so they carry the most weight in clinical decisions.



What is a practice guideline?

- Based on the highest available level of evidence.
- Built on systematic review and meta-analysis.
- Designed to support decision-making in patient care.
- Constantly updated as new evidence appears.

In one phrase: doing the right thing. A guideline turns the best available evidence into a clear recommendation you can act on at the bedside.

Levels of evidence

LEVEL	EVIDENCE BASED ON
Ia	Meta-analyses of randomised controlled trials
Ib	At least one randomised controlled trial
IIa	At least one well-planned, non-randomised controlled study
IIb	At least one well-planned, non-randomised, non-controlled study (e.g. cohort)
III	Well-planned, non-experimental descriptive studies (e.g. case-control)
IV	Reports or opinions of expert committees / recognised authorities

Strength of recommendation

GRADE	SUPPORTING EVIDENCE
A	Level Ia or Ib — first class from the clinical point of view
B	Level IIa, IIb or III — second class from the clinical point of view
C	Level IV — third class from the clinical point of view

ASKING AND SEARCHING

The PICO format

Articulate your need for information before you touch the database.

P	Patient	Who is the patient, and what is the problem?
I	Intervention	Which treatment or exposure are you considering?
C	Comparator	What is the alternative you compare it against?
O	Outcome	What result are you trying to achieve?

Searching PubMed for a guideline

Database	Medline, via PubMed
Search terms	MeSH terms built from the P and I of your PICO
Filters	Practice guideline · Humans · English

Filter to **practice guideline** because it sits at the top of the evidence pyramid and gives you the strongest, most current recommendation.

Is this guideline worth following?

Four questions to put to every guideline you retrieve.

Who wrote it?

A recognised professional body or national agency, with a named panel, a stated method and declared conflicts of interest.

How recent is it?

Check the publication year and whether a newer update or focused amendment exists — anticoagulation guidance changed substantially once the NOACs entered practice.

Is the evidence graded?

Each recommendation should carry a strength and a quality-of-evidence rating. A strong recommendation resting on low-quality evidence is a flag, not a fact.

Does it apply to my patient?

Compare the guideline population with your patient, and check that the recommended drug, its monitoring and its antidote are actually available to you.

Good is not the same as applicable. Drug availability, monitoring capacity and cost decide whether a recommendation can be followed in Egyptian practice.

FINDING ANSWERS INSIDE A GUIDELINE

Guidelines are long. These five entry points take you from a two-hundred-page document to the sentence that answers your question.

1 Start with the index

Guidelines are long. The table of contents takes you straight to the section that answers your question — look for the recommendation heading that matches your intervention.

Index:

1. Introduction 2023

1.1. Methodology and Evidence Review 2023
1.2. Organization of the Writing Group 2024
1.3. Document Review and Approval 2024
1.4. Scope of the Focused Update 2024
2. Lower Extremity PAD 2024

2.5. Diagnostic Methods 2024
2.5.1. Recommendations for Ankle-Brachial Index, Toe-Brachial Index, and Segmental Pressure Examination 2024
2.6. Treatment 2025
2.6.1.4. RECOMMENDATIONS FOR SMOKING CESSATION 2025
2.6.1.6. RECOMMENDATIONS FOR ANTIPLATELET AND ANTITHROMBOTIC DRUGS ..2025
2.6.3. Recommendations for Critical Limb Ischemia: Endovascular and Open Surgical Treatment for Limb Salvage 2027
5. Aneurysm of the Abdominal Aorta, Its Branch Vessels, and the Lower Extremities ..2028
5.2.8.1. RECOMMENDATIONS FOR MANAGEMENT OVERVIEW 2028
References 2031

2 Read the summary of recommendations

Each recommendation is graded by class and carries a level of evidence. Report both: “beta-blockers, level of evidence B”.

Summary of Recommendations:

6.2.1. Pharmacologic Management—Recommendations

CLASS I

1. Beta-blocking drugs are recommended for the treatment of symptoms (angina or dyspnea) in adult patients with obstructive or nonobstructive HCM but should be used with caution in patients with sinus bradycardia or severe conduction disease (3,9,10,134,137,229–236). *(Level of Evidence: B)*
2. If low doses of beta-blocking drugs are ineffective for controlling symptoms (angina or dyspnea) in patients with HCM, it is useful to titrate the dose to a resting heart rate of less than 60 to 65 bpm (up to generally accepted and recommended maximum doses of these drugs) (3,10,137, 229–236). *(Level of Evidence: B)*
3. Verapamil therapy (starting in low doses and titrating up to 480 mg/d) is recommended for the treatment of symptoms (angina or dyspnea) in patients with obstructive or nonobstructive HCM who do not respond to beta-blocking drugs or who have side effects or contraindications to beta-blocking drugs. However, verapamil should be used with caution in patients with high gradients, advanced heart failure, or sinus bradycardia (10,134,137,237–241). *(Level of Evidence: B)*
4. Intravenous phenylephrine (or another pure vasoconstricting agent) is recommended for the treatment of acute hypotension in patients with obstructive HCM who do not respond to fluid administration (137,242–244). *(Level of Evidence: B)*

3 Read the tables

Tables set old and new recommendations side by side with the evidence level in each cell — useful when the document is a focused update and you need to see exactly what changed.

Tables:

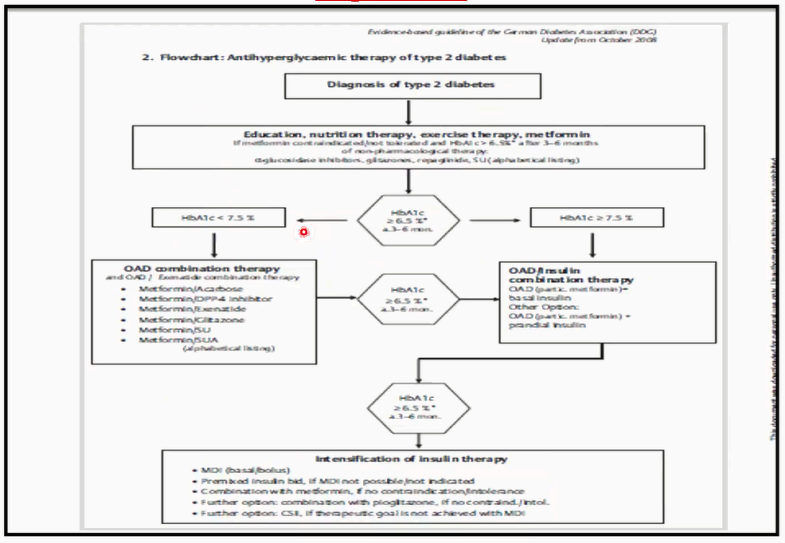
Table 2. Updates to Section 2.3.1. Endocarditis Prophylaxis

2006 VHD Guideline Recommendations	2008 VHD Focused Update Recommendations	Comments
Class I	Class IIa	
1. Prophylaxis against infective endocarditis is recommended for the following patients:	1. Prophylaxis against infective endocarditis is reasonable for the following patients at highest risk for adverse outcomes from infective endocarditis who undergo dental procedures that involve manipulation of either gingival tissue or the periapical region of teeth or perforation of the oral mucosa :	Modified recommendation (changed class of recommendation from I to IIa, changed text). There are no Class I recommendations for infective endocarditis prophylaxis.
· Patients with prosthetic heart valves and patients with a history of infective endocarditis. *(Level of Evidence: C)*		
· Patients who have complex cyanotic congenital heart disease (e.g., single–ventricle states, transposition of the great arteries, tetralogy of Fallot). *(Level of Evidence: C)*	· Patients with prosthetic cardiac valves or prosthetic material used for cardiac valve repair. *(Level of Evidence: B)*	

4 Follow the algorithm

Flowcharts turn the recommendations into a decision path. Start at the diagnosis and follow the branches that match your patient's values.

Algorithm:



5 Scan the headings and narrative

The narrative carries the detail the summary leaves out — doses, routes and monitoring, each with its evidence grade in brackets.



SESSION 1 — THROMBOTIC DISEASE

Case 1 · Travel-Related Thrombosis

A 54-year-old woman attended the clinic two days before a planned 13-hour flight. She had undergone major abdominal surgery three weeks earlier and had made a good recovery. She is otherwise well and takes no regular medication. She read online that passengers should drink plenty of water and take aspirin before long flights to avoid travel-related thrombosis. She also asks how long after the flight she would remain at risk of developing a clot.

STEP 1 — ARTICULATE THE NEED: PICO

P	**Patient**	A 54-year-old woman, major surgery three weeks ago, a 13-hour flight, at risk of travel-related thrombosis
I	**Intervention**	
C	**Comparator**	
O	**Outcome**	Prevention (prophylaxis) of travel-related thrombosis

STEP 2 — SEARCH QUESTION

What is the best prophylactic drug for a 54-year-old woman who underwent major surgery three weeks ago and is travelling for 13 hours?

STEP 3 — SEARCH STRATEGY

Database	Medline (via PubMed)
MeSH terms	“Thrombosis” AND “Travel”
Filters	Practice guideline · Humans · English

STEP 4 — THE ARTICLE YOU SELECTED

Title	
Type, and why	Practice guideline — it sits at the top of the evidence pyramid and gives the strongest, most current recommendation.

STEP 5 — FIND THE ANSWERS IN THE GUIDELINE, AND STATE THE LEVEL OF EVIDENCE

1 **What is the magnitude of the risk of venous thromboembolism associated with long-distance travel, and for how long after the journey may a VTE still be attributed to travel?**

Level of evidence: _____

2 **What advice should be given to travellers regarding hydration, mobility, compression stockings and drug prophylaxis — and which travellers are identified as being at highest risk?**

Level of evidence: _____

3 **Using the guideline's risk-group / duration table, which risk category does this patient fall into, and what specific prophylaxis is advised for her 13-hour flight?**

Level of evidence: _____

Case 2 · Cancer-Related DVT

A 61-year-old man with locally advanced pancreatic adenocarcinoma receiving systemic chemotherapy presented with a painful, swollen left lower limb of two days' duration. He had no bleeding, a platelet count of 210 × 10^9/l, a creatinine clearance of 85 ml/min and normal liver function. Doppler ultrasound confirmed an acute proximal (femoropopliteal) deep vein thrombosis.

STEP 1 — ARTICULATE THE NEED: PICO

P	**Patient**	A 61-year-old man with locally advanced pancreatic adenocarcinoma and an acute proximal DVT
I	**Intervention**	Low molecular weight heparin
C	**Comparator**	Oral anticoagulant
O	**Outcome**	Treatment and prevention of recurrent thromboembolism (secondary prevention)

STEP 2 — SEARCH QUESTION

What is the best treatment to treat and prevent recurrence of thromboembolism in a 61-year-old man with locally advanced pancreatic adenocarcinoma and an acute proximal DVT?

STEP 3 — SEARCH STRATEGY

Database	Medline (via PubMed)
MeSH terms	“Venous thromboembolism” OR “Neoplasms” AND “Venous thromboembolism”
Filters	Practice guideline · Humans · English

STEP 4 — THE ARTICLE YOU SELECTED

Title	
Type, and why	Practice guideline — it sits at the top of the evidence pyramid and gives the strongest, most current recommendation.

STEP 5 — FIND THE ANSWERS IN THE GUIDELINE, AND STATE THE LEVEL OF EVIDENCE

1 **What workup and imaging does the guideline recommend for a patient with clinical suspicion of DVT? What should be done if the initial ultrasound is negative or indeterminate but suspicion persists, and may anticoagulation be started before imaging is completed?**

Level of evidence: _____

2 **Which anticoagulant regimen would you prescribe, with the exact dose? How does the site of his tumour influence the choice between a DOAC and LMWH, and which agents carry category 1 evidence?**

Level of evidence: _____

3 **For how long should he be anticoagulated, and which patient factors should guide the choice of regimen and the baseline and follow-up laboratory monitoring?**

Level of evidence: _____

Case 3 · Peripheral Arterial Disease

A 68-year-old man attended the vascular clinic with a nine-month history of cramping pain in both calves brought on after walking about 150 metres and relieved by rest, now limiting his daily activity. He is a current smoker (40 pack-years) with type 2 diabetes and hypertension. Pedal pulses were diminished and the ankle–brachial index was 0.62 on the right and 0.68 on the left, confirming symptomatic peripheral artery disease. His LDL-C was 132 mg/dl and he was taking aspirin 81 mg daily alone, with no lipid-lowering therapy and no history of bleeding or revascularisation.

STEP 1 — ARTICULATE THE NEED: PICO

P	**Patient**	A 68-year-old man with symptomatic peripheral artery disease and functionally limiting claudication (ABI 0.62 / 0.68), a current smoker with diabetes
I	**Intervention**	Low-dose rivaroxaban plus low-dose aspirin
C	**Comparator**	Aspirin alone
O	**Outcome**	Prevention of major adverse cardiovascular and limb events (MACE and MALE) — secondary prevention

STEP 2 — SEARCH QUESTION

What is the best antithrombotic therapy to prevent major adverse cardiovascular and limb events in a 68-year-old man with symptomatic peripheral artery disease and functionally limiting claudication?

STEP 3 — SEARCH STRATEGY

Database	Medline (via PubMed)
MeSH terms	“Peripheral arterial disease” AND “Platelet aggregation inhibitors”
Filters	Practice guideline · Humans · English

STEP 4 — THE ARTICLE YOU SELECTED

Title	
Type, and why	Practice guideline — it sits at the top of the evidence pyramid and gives the strongest, most current recommendation.

STEP 5 — FIND THE ANSWERS IN THE GUIDELINE, AND STATE THE LEVEL OF EVIDENCE

1 **What antithrombotic regimen should this patient be prescribed, at what dose, and what documented medical, patient or system reasons would legitimately exempt a patient from this measure?**

Level of evidence: _____

2 **What lipid-lowering therapy is expected for him, what LDL-C reduction is targeted, and how is "high-intensity statin therapy" defined in this document?**

Level of evidence: _____

3 **What exercise-based therapy should be offered for his functionally limiting claudication, which alternative forms of exercise are supported, and what is the standing of simple advice to "go out and walk"?**

Level of evidence: _____

Case 4 · Heparin-Induced Thrombocytopenia

A 72-year-old woman was admitted for surgical fixation of a hip fracture and received prophylactic unfractionated heparin postoperatively. On day 8 her platelet count had fallen from 245 × 10^9/l to 62 × 10^9/l (a 75% fall), and she developed swelling of the right calf; Doppler ultrasound confirmed an acute proximal deep vein thrombosis. There was no bleeding and no other obvious cause of thrombocytopenia. Her 4T score was 7 and her creatinine clearance was 22 ml/min. Anti-PF4 antibody testing was sent but results would not be available for 48 hours.

STEP 1 — ARTICULATE THE NEED: PICO

P	**Patient**	A 72-year-old woman on unfractionated heparin after hip fracture surgery, with a 75% fall in platelet count, acute proximal DVT, a 4T score of 7 and severe renal impairment
I	**Intervention**	Argatroban
C	**Comparator**	Danaparoid / fondaparinux / DOAC
O	**Outcome**	Treatment of heparin-induced thrombocytopenia and its thrombosis, and prevention of new thrombotic events

STEP 2 — SEARCH QUESTION

What is the best non-heparin anticoagulant to treat heparin-induced thrombocytopenia and prevent new thrombosis in a 72-year-old woman with a high 4T score, acute proximal DVT and severe renal impairment?

STEP 3 — SEARCH STRATEGY

Database	Medline (via PubMed)
MeSH terms	“Heparin” AND “Thrombocytopenia”
Filters	Practice guideline · Humans · English

STEP 4 — THE ARTICLE YOU SELECTED

Title	
Type, and why	Practice guideline — it sits at the top of the evidence pyramid and gives the strongest, most current recommendation.

STEP 5 — FIND THE ANSWERS IN THE GUIDELINE, AND STATE THE LEVEL OF EVIDENCE

1 **How is the clinical probability of heparin-induced thrombocytopenia classified, which patients need anti-PF4 antibody testing, and what should be done about her heparin while the laboratory results are awaited?**

Level of evidence: _____

2 **Which non-heparin anticoagulants may be used in the acute phase? How is the choice individualised for a stable patient, an unstable or bleeding-risk patient, severe HIT, severe renal impairment and severe hepatic impairment?**

Level of evidence: _____

3 **For the agent indicated in her case, what initial dose is proposed, how should it be monitored biologically, and at what point may a vitamin K antagonist be introduced?**

Level of evidence: _____

SESSION 2 — ANEMIA

Case 1 · Iron-Deficiency Anemia

A 58-year-old man presented complaining of fatigue and exertional shortness of breath. His Hb was 8.2 g/dl, MCV 72 fl and serum ferritin 9 µg/l. He was diagnosed with iron-deficiency anemia and started on one tablet of ferrous sulfate 200 mg daily, but stopped after 10 days because of severe nausea and constipation. Two weeks later his Hb was still 8.4 g/dl.

STEP 1 — ARTICULATE THE NEED: PICO

P	**Patient**	A 58-year-old man with fatigue and exertional dyspnoea; Hb 8.2 g/dl, MCV 72 fl, ferritin 9 µg/l
I	**Intervention**	Ferrous sulfate 200 mg daily
C	**Comparator**	Parenteral iron
O	**Outcome**	Treatment of iron-deficiency anemia after failed oral therapy

STEP 2 — SEARCH QUESTION

What is the best treatment of iron-deficiency anemia in a 58-year-old man with fatigue and exertional shortness of breath in whom oral treatment has failed?

STEP 3 — SEARCH STRATEGY

Database	Medline (via PubMed)
MeSH terms	“Anemia, iron-deficiency”
Filters	Practice guideline · Humans · English

STEP 4 — THE ARTICLE YOU SELECTED

Title	
Type, and why	Practice guideline — it sits at the top of the evidence pyramid and gives the strongest, most current recommendation.

STEP 5 — FIND THE ANSWERS IN THE GUIDELINE, AND STATE THE LEVEL OF EVIDENCE

1 **What is the recommended initial oral iron regimen for this patient, and what should be done if he cannot tolerate it?**

Level of evidence: _____

2 **In this patient, when should parenteral (intravenous) iron be considered instead of oral?**

Level of evidence: _____

3 **How should the response to iron replacement be monitored, and what rise in Hb is considered an adequate response?**

Level of evidence: _____

4 **What is the recommended long-term treatment to prevent recurrence, and its level of evidence?**

Level of evidence: _____

Case 2 · Iron Deficiency in Chronic Kidney Disease

A 58-year-old woman with chronic kidney disease stage G4 (eGFR 21 ml/min/1.73 m^2), not receiving dialysis, attended the nephrology clinic complaining of fatigue and poor exercise tolerance. Her Hb was 9.6 g/dl, serum ferritin 85 ng/ml and transferrin saturation (TSAT) 19%. There was no overt bleeding and no active infection. She lives far from the hospital and prefers to avoid frequent visits.

STEP 1 — ARTICULATE THE NEED: PICO

P	**Patient**	A 58-year-old woman with CKD stage G4 (eGFR 21 ml/min/1.73 m^2), Hb 9.6 g/dl, ferritin 85 ng/ml, TSAT 19%
I	**Intervention**	Oral iron therapy
C	**Comparator**	Parenteral iron therapy
O	**Outcome**	Correction of anemia and iron deficiency in non-dialysis CKD

STEP 2 — SEARCH QUESTION

What is the best treatment of anemia and iron deficiency in a 58-year-old woman with CKD stage G4 who is not on dialysis?

STEP 3 — SEARCH STRATEGY

Database	Medline (via PubMed)
MeSH terms	“Renal insufficiency, chronic” AND “Anemia, iron-deficiency”
Filters	Practice guideline · Humans · English

STEP 4 — THE ARTICLE YOU SELECTED

Title	
Type, and why	Practice guideline — it sits at the top of the evidence pyramid and gives the strongest, most current recommendation.

STEP 5 — FIND THE ANSWERS IN THE GUIDELINE, AND STATE THE LEVEL OF EVIDENCE

1 **According to the guideline, is this patient anemic? What tests screen for anemia and iron deficiency in CKD, and do her ferritin and TSAT values meet the threshold to start iron?**

Level of evidence: _____

2 **Which route of iron therapy should she receive, what should be done if she does not respond or cannot tolerate it, how often should iron parameters be checked, and when should iron be withheld?**

Level of evidence: _____

3 **Three months later her iron stores are replete (ferritin 240 ng/ml, TSAT 28%) but her Hb is 9.0 g/dl and she remains symptomatic. At what Hb level should ESA therapy be started in non-dialysis CKD, and what Hb target should be maintained once she is on an ESA?**

Level of evidence: _____

Case 3 · Macrocytic Anemia

A 67-year-old man presented complaining of progressive fatigue over four months, a sore red tongue, and tingling and numbness in both feet. He has treated hypothyroidism. His Hb was 8.9 g/dl with an MCV of 114 fl; the blood film showed oval macrocytes and hypersegmented neutrophils. Serum cobalamin was 110 pmol/l and serum folate was normal. There was no history of gastrointestinal surgery and he is not a vegetarian.

STEP 1 — ARTICULATE THE NEED: PICO

P	**Patient**	An adult with anemia, neuropathy and glossitis, suspected of having pernicious anemia
I	**Intervention**	Vitamin B12 therapy
C	**Comparator**	Folate therapy
O	**Outcome**	Treatment of macrocytic anemia due to cobalamin deficiency

STEP 2 — SEARCH QUESTION

What is the best treatment of macrocytic anemia in an adult with anemia, neuropathy or glossitis suspected of having pernicious anemia?

STEP 3 — SEARCH STRATEGY

Database	Medline (via PubMed)
MeSH terms	“Anemia, pernicious” OR “Vitamin B12 deficiency”
Filters	Practice guideline · Humans · English

STEP 4 — THE ARTICLE YOU SELECTED

Title	
Type, and why	Practice guideline — it sits at the top of the evidence pyramid and gives the strongest, most current recommendation.

STEP 5 — FIND THE ANSWERS IN THE GUIDELINE, AND STATE THE LEVEL OF EVIDENCE

1 **Which blood film findings support this diagnosis, which two assays should be requested together and why, and what serum cobalamin cut-off is taken as evidence of deficiency?**

Level of evidence: _____

2 **Which antibody test should be requested to establish the underlying cause, and does a normal serum cobalamin remove the need for it? Is testing for gastric parietal cell antibody recommended?**

Level of evidence: _____

3 **Outline the treatment regimen given his neurological symptoms — initial therapy and maintenance — and state when a haematological response should be expected and what a suboptimal response may indicate.**

Level of evidence: _____

SESSION 3 — FURTHER PRACTICE

Case 1 · Imported Malaria

A 23-year-old man who arrived from Nigeria 15 days ago presented to the physician's office with a history of low-grade fever that started four days ago, with no other symptoms. Blood smear examination confirmed malaria, and the physician prescribed chloroquine.

STEP 1 — ARTICULATE THE NEED: PICO

P	**Patient**	A 23-year-old man who arrived from Nigeria 15 days ago, low-grade fever for four days, confirmed malaria
I	**Intervention**	Chloroquine
C	**Comparator**	Proguanil, or any other antimalarial drug
O	**Outcome**	Treatment of malaria

STEP 2 — SEARCH QUESTION

What is the best treatment of malaria in a 23-year-old man who arrived from Nigeria 15 days ago?

STEP 3 — SEARCH STRATEGY

Database	Medline (via PubMed)
MeSH terms	“Malaria”
Filters	Practice guideline · Humans · English

STEP 4 — THE ARTICLE YOU SELECTED

Title	
Type, and why	Practice guideline — it sits at the top of the evidence pyramid and gives the strongest, most current recommendation.

STEP 5 — FIND THE ANSWERS IN THE GUIDELINE, AND STATE THE LEVEL OF EVIDENCE

1 **Which species is most likely given travel from this region, and what does the guideline recommend as first-line treatment for uncomplicated falciparum malaria in a returning traveller? Is the chloroquine he was prescribed appropriate here, and why?**

Level of evidence: _____

2 **What clinical and laboratory criteria define severe malaria, and which parenteral agent, dose and route does the guideline recommend once those criteria are met?**

Level of evidence: _____

3 **What does the guideline advise about eradicating dormant hepatic forms and preventing relapse, and what must be checked before an 8-aminoquinoline is prescribed?**

Level of evidence: _____

Case 2 · Dabigatran Before Surgery

A 32-year-old woman was admitted to hospital for a surgical procedure. During pre-operative assessment she reported taking an oral anticoagulant (dabigatran) for non-valvular atrial fibrillation. The anaesthetist wants to know when to stop the drug before surgery. Her creatinine clearance was 30 ml/min (normal for a healthy woman: 88–128 ml/min).

STEP 1 — ARTICULATE THE NEED: PICO

P	**Patient**	A 32-year-old woman on dabigatran for non-valvular atrial fibrillation, due for a surgical procedure, creatinine clearance 30 ml/min
I	**Intervention**	Stopping dabigatran before surgery
C	**Comparator**	Continuing dabigatran
O	**Outcome**	Avoiding peri-operative bleeding while preventing thromboembolism

STEP 2 — SEARCH QUESTION

What is the best time to stop dabigatran before a surgical procedure in a 32-year-old woman with a creatinine clearance of 30 ml/min?

STEP 3 — SEARCH STRATEGY

Database	Medline (via PubMed)
MeSH terms	“Dabigatran” AND “Surgical procedures, operative”
Filters	Practice guideline · Humans · English

STEP 4 — THE ARTICLE YOU SELECTED

Title	
Type, and why	Practice guideline — it sits at the top of the evidence pyramid and gives the strongest, most current recommendation.

STEP 5 — FIND THE ANSWERS IN THE GUIDELINE, AND STATE THE LEVEL OF EVIDENCE

1 **How long before surgery should dabigatran be stopped, and how does that interval change with creatinine clearance and with the bleeding risk of the planned procedure? What interval applies to this patient?**

Level of evidence: _____

2 **Is bridging with heparin or LMWH recommended when a NOAC is interrupted, and when may dabigatran be restarted after surgery?**

Level of evidence: _____

3 **If she needed urgent surgery instead, what reversal strategy does the guideline recommend, and which laboratory tests indicate a residual anticoagulant effect?**

Level of evidence: _____